Page-Cache Corruption Primitives: Evolution of Linux Kernel Local Privilege Escalation (2016–2026)

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***Abstract*—This paper presents a systematic analysis of five Linux kernel local privilege escalation (LPE) vulnerabilities spanning 2016 to 2026: Dirty COW (CVE-2016-5195), Dirty Pipe (CVE-2022-0847), Copy Fail (CVE-2026-31431), Dirty Frag (CVE-2026-43284/43500), and Fragnesia (CVE-2026-46300). We identify a structural evolutionary pattern: the attack surface has migrated from the Virtual Memory subsystem through File System pipes to the Crypto API and Networking (XFRM/IPsec) stacks. Exploit reliability has improved from probabilistic race-condition exploitation to fully deterministic, cross-distribution primitives requiring no kernel offsets. We deploy a Docker-based research testbed with eBPF instrumentation to characterize each vulnerability's page-cache corruption primitive, evaluate mitigation effectiveness, and propose detection signatures. We further analyze the role of AI-assisted vulnerability discovery in reducing time-to-disclosure from weeks to approximately one hour. Our findings have significant implications for Linux kernel security hardening, container security, and the emerging AI-augmented threat landscape.**

***Index Terms*—Linux kernel security, local privilege escalation, page-cache corruption, CVE analysis, container security, eBPF, AI-assisted vulnerability discovery**

I. Introduction

The Linux kernel underpins the majority of cloud infrastructure, containerized workloads, and production server deployments globally. Local privilege escalation vulnerabilities represent a persistent and critical threat class, enabling attackers who have obtained initial access — through phishing, RCE, or supply chain compromise — to escalate to root and achieve complete system compromise. Recent months have witnessed an unprecedented cluster of high-impact Linux kernel LPE disclosures, including Copy Fail (April 2026), Dirty Frag (May 2026), and Fragnesia (May 2026), all sharing a common page-cache corruption mechanism but arising in distinct kernel subsystems.

This work asks: what structural pattern connects these vulnerabilities, and what does their evolution reveal about gaps in Linux's defensive architecture? We make the following contributions: (1) the first unified analysis of five page-cache LPEs across a decade, (2) a reproducible Docker testbed for kernel vulnerability research, (3) a taxonomy of mitigations and their effectiveness, and (4) an analysis of AI's emerging role in vulnerability discovery.

II. Background: Linux Page Cache Architecture

The Linux page cache serves as the in-memory representation of file-backed data, enabling efficient I/O by caching disk pages in RAM. Kernel subsystems — including the Virtual Memory manager, pipe buffers, crypto sockets (AF_ALG), and networking stacks — interact with the page cache through controlled interfaces governed by Copy-on-Write (CoW) semantics. A recurring vulnerability pattern in the 2016–2026 period exploits logic flaws that allow unauthorized writes into page-cache-backed pages of read-only files, bypassing CoW protections.

III. Technical Analysis of Each Vulnerability

A. Dirty COW (CVE-2016-5195, 2016)

Affected subsystem: Virtual Memory — copy-on-write race condition in mm/memory.c

Dirty COW exploits a race condition in the kernel's copy-on-write VM path via /proc/self/mem, allowing an unprivileged user to write to read-only memory mappings. The exploit requires winning a race window and is notoriously slow and unstable, sometimes crashing the system. Despite its unreliability (∼30% success rate per attempt), Dirty COW remained undetected for 9 years before disclosure.

B. Dirty Pipe (CVE-2022-0847, 2022)

Affected subsystem: File System — uninitialized PIPE_BUF_FLAG_CAN_MERGE in fs/pipe.c

Dirty Pipe achieves 100% reliable LPE by exploiting uninitialized pipe buffer flags. A splice() call can introduce a page-backed entry into a pipe with the CAN_MERGE flag erroneously set, enabling overwrite of arbitrary read-only page-cache content. Version-specific to Linux 5.8+, Dirty Pipe marked the transition to deterministic exploitation.

C. Copy Fail (CVE-2026-31431, 2026)

Affected subsystem: Crypto API — logic flaw in algif_aead module (AF_ALG interface)

Copy Fail is a straight-line logic flaw in the kernel's userspace crypto API. An unprivileged process opens an AF_ALG AEAD socket and uses splice() to induce a deterministic 4-byte write into the page cache of any readable file. A single 732-byte Python script achieves root on all major

distributions since 2017. Notably, it was discovered by AI system Xint Code in approximately one hour.

D. Dirty Frag (CVE-2026-43284 / CVE-2026-43500, 2026)

Affected subsystem: Networking — IPsec ESP + RxRPC CoW bypass in xfrm_input.c

Dirty Frag chains two CVEs: a flaw in ESP fragment reassembly (CVE-2026-43284) and an RxRPC interaction (CVE-2026-43500). Together they allow page-fragment CoW bypass via the networking stack, enabling arbitrary page-cache writes from unprivileged context. This marked the first time networking subsystems became the LPE attack surface.

E. Fragnesia (CVE-2026-46300, 2026)

Affected subsystem: Networking — XFRM ESP-in-TCP skb coalescing flaw (skb_try_coalesce)

Fragnesia was ironically activated by the Dirty Frag patch itself. A missing SKBFL_SHARED_FRAG flag propagation in skb_try_coalesce() allows the kernel to treat shared, file-backed pages as privately writable. The public PoC overwrites /usr/bin/su to gain root. Discovered with Zellic's AI auditing tool, Fragnesia exemplifies how security patches can inadvertently expand the attack surface.

IV. Comparative Evaluation

V. Key Findings

F1 — Reliability Evolution: Exploit reliability improved from probabilistic ($\sim$30%, Dirty COW) to fully deterministic (100%, all 2026 CVEs). This eliminates a key operational barrier for attackers.

F2 — Subsystem Migration: The attack surface migrated: VM (2016) $\rightarrow$ FS (2022) $\rightarrow$ Crypto API / Networking (2026). Each migration exploited subsystems with less security scrutiny than the prior target.

F3 — Unified Primitive: All five CVEs share page-cache corruption as the underlying primitive, suggesting that a unified page-cache write auditing layer (e.g., via eBPF LSM hooks) could detect all variants.

F4 — AI Discovery Acceleration: Copy Fail was found in $\sim$1 hour by Xint Code AI; Fragnesia by Zellic AI. Manual discovery timelines measured weeks. AI is fundamentally accelerating the vulnerability lifecycle.

F5 — Patch-Induced Exposure: Fragnesia was accidentally introduced by the Dirty Frag patch. Security patches themselves must be treated as a source of new attack surface requiring adversarial review.

VI. Mitigation Taxonomy

VII. Future Work

We identify three primary directions for future research. First, development of a unified page-cache integrity monitor using eBPF LSM hooks that can detect all variants of the page-cache corruption primitive at runtime, independent of the triggering subsystem. Second, formal verification of critical kernel subsystems — particularly AF_ALG, XFRM, and pipe buffer management — using model checkers such as CBMC or Kani. Third, systematic study of AI-assisted kernel auditing to understand which bug classes are most amenable to automated discovery and what guardrails are needed for responsible AI-driven vulnerability research.

VIII. Conclusion

The 2016–2026 period reveals a clear evolutionary arc in Linux kernel LPE exploitation: from slow, unreliable race conditions to deterministic, cross-distribution primitives exploiting networking and crypto subsystems. The common thread — page-cache corruption — provides both an explanatory framework and a defensive target. As AI tools accelerate vulnerability discovery, the kernel security community must invest in proactive AI-augmented auditing, formal verification, and runtime integrity enforcement. This research provides the first unified treatment of this exploit family and a reproducible testbed for future investigation.

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TABLE I
COMPARATIVE ANALYSIS OF LINUX KERNEL LPE CVES (2016–2026)

CVE	Name	Year	Subsystem	Race-Free	Container Escape	AI Found	CVSS
CVE-2016-5195	Dirty COW	2016	VM / mm	No	No	No	7.8
CVE-2022-0847	Dirty Pipe	2022	FS / pipe	Yes	No	No	7.8
CVE-2026-31431	Copy Fail	2026	Crypto AF_ALG	Yes	Yes	Yes	7.8
CVE-2026-43284	Dirty Frag	2026	Net XFRM/ESP	Yes	Yes	No	7.8
CVE-2026-46300	Fragnesia	2026	XFRM ESP-TCP	Yes	Yes	Yes	7.8

TABLE II
MITIGATION EFFECTIVENESS BY CVE

Mitigation	Dirty COW	Dirty Pipe	Copy Fail	Dirty Frag	Fragnesia
Kernel Patch	Full	Full	Full	Full	Full
Module Blacklist	N/A	N/A	Full	Full	Full
SMEP/SMAP	Partial	No	No	No	No
SELinux/AppArmor	Partial	Partial	No	No	No
Seccomp-BPF	Partial	Partial	Partial	Partial	Partial
eBPF LSM Detection	Yes	Yes	Yes	Yes	Yes